UAV patrolling for **wildfire** monitoring by a dynamic voronoi tessellation on satellite data

A Giuseppi, R Germanà, F Fiorini, F Delli Priscoli… - Drones, 2021 - mdpi.com

… The **prediction** of the **wildfire propagation** is a crucial functionality needed to enable any kind of proactive measure. For this reason, several models [22,23,24,25] have been proposed in …

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Do thinning and/or burning treatments in western USA ponderosa or Jeffrey pine-dominated forests help restore natural fire **behavior**?

PZ Fulé, JE Crouse, JP Roccaforte, EL Kalies - … Ecology and **Management**, 2012 - Elsevier

… and burning treatments on restoring fire **behavior** attributes in western USA pine forests. … of **wildfire behavior** in treated stands have also been reported. Using a systematic **review** …

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A **review** of fire interactions and mass fires

MA Finney, SS McAllister - Journal of Combustion, 2011 - Wiley Online Library

… , including **wildland fire**, structural fire, as well as combustion engineering and fluid **dynamics**, … on interactions is directly relevant to **wildfire management** activities and tactics but has not …

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Fire **behavior** in masticated fuels: a **review**

JK Kreye, NW Brewer, P Morgan, JM Varner… - … and **Management**, 2014 - Elsevier

… Managing landscapes dependent on fire as an ecological process, while also mitigating the negative impacts of **wildfire** to people and property is challenging. In many fire-prone forest …

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Application of **wildfire** risk assessment results to **wildfire** response planning in the southern Sierra Nevada, California, USA

MP Thompson, P Bowden, A Brough, JH Scott… - Forests, 2016 - mdpi.com

… We **review** general … **wildfire** hazard across the landscape at a pixel size of 180 m (3 ha per pixel). FSim is a comprehensive fire occurrence, growth, **behavior**, and suppression **simulation** …

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A GIS-supported **model** for the **simulation** of the spatial structure of **wildland fire**, Cass Basin, New Zealand

GLW Perry, AD Sparrow… - Journal of Applied Ecology, 1999 - Wiley Online Library

… **spread model** to New Zealand (NZ) ecosystems, and secondly, to assess the role GIS may play in fire **spread prediction**… to the data from the **survey** plots. Before dissection each whole (…

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Cooperative **forest fire** surveillance using a team of small unmanned air vehicles

DW Casbeer, DB Kingston, RW Beard… - International journal of …, 2006 - Taylor & Francis

… in **simulation** using a six degree-of-freedom dynamic **model** for the UAV and a numerical **propagation model** for the **forest fire**. … Let one UAV be assigned to **survey** the upper half of the …

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Self-autonomous wireless sensor nodes with wind energy harvesting for remote sensing of wind-driven **wildfire spread**

YK Tan, SK Panda - IEEE Transactions on Instrumentation and …, 2011 - ieeexplore.ieee.org

… coordinating the **review** process for … **Prediction** of the **Wildfire Spread** Based on the Wind Speed In the remote sensing of the **wildfire spread** condition, the **prediction** of the fire **behavior** …

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Coupling diagnostic and prognostic models to a dynamic data driven **forest fire spread prediction** system

C Brun, T Margalef, A Cortés - Procedia Computer Science, 2013 - Elsevier

… Finally, when considering the scheme where both prognostic and diagnostic models are coupled to the **forest fire spread model**, the **prediction spread** errors on average denote a great …

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Wildfire Impacts on Security of Electric Power Systems: A **Survey** of Risk Identification and Mitigation Approaches

R Shah, N Amjady, NSB Shakhawat, MU Aslam… - IEEE …, 2024 - ieeexplore.ieee.org

… The work in [44] has used a **wildfire prediction model** using … and recovery including **wildfire prediction** and network … to estimate the **wildfire dynamics** and **propagation** direction and its …

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Comparisons between **wildfire** and forest harvesting and their implications in forest **management**

DJ McRae, LC Duchesne, B Freedman… - Environmental …, 2001 - cdnsciencepub.com

… **review** we summarize empirical evidence that illustrates disparities between forest harvesting and **wildfire**… of disturbance by **wildfire** offer valuable insights into the ecological **dynamics** of …

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Wildland fire detection and monitoring using a drone-collected rgb/ir image dataset

X Chen, B Hopkins, H Wang, L O'Neill, F Afghah… - IEEE …, 2022 - ieeexplore.ieee.org

… leave uncertainties about a **wildfire**’s extent, **behavior**, and … of drones in remote **wildfire management** operations [12], [13]… Table 1 provides a **review** of existing fire datasets. As shown …

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Machine Learning based **Wildfire** Area Estimation Leveraging Weather **Forecast** Data

S Sultania, R Sonawane, P Kanikar - Int. J. Inf. Technol. Comput …, 2025 - mecs-press.org

… **wildfire prediction** to be used in a literature **review** as follows: … , improving **prediction** accuracy for **wildfire spread**. Bayesian … the inherent uncertainties of **wildfire prediction**. Reference [12…

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Forest fire spread prediction using deep learning

F Khennou, J Ghaoui, MA Akhloufi - Geospatial informatics XI, 2021 - spiedigitallibrary.org

… , a deep learning **model** based on U-Net, past wildfires events and weather data. Our approach uses an intelligent **model** to study **forest fire spread** over a period of 24 hours. The **model** …

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Phoenix: development and application of a **bushfire** risk **management** tool

K Tolhurst, B Shields, D Chong - … of Emergency **Management** …, 2008 - search.informit.org

… measures for monitoring and **review** make up the risk **management** process as outlined in … the **Bushfire** Risk **Management Model** encapsulate the complex interaction of ignition, **spread**, …

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A **review** of challenges to determining and demonstrating efficiency of large fire **management**

MP Thompson, FR y Silva, DE Calkin… - … Journal of **Wildland Fire**, 2017 - CSIRO Publishing

Characterising the impacts of **wildland fire** and fire suppression is critical information for fire **management** decision-making. Here, we focus on decisions related to the rare larger and …

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Predicting crown fire **behavior** to support **forest fire management** decision-making

MG Cruz, ME Alexander, RH Wakimoto - … conference on **forest fire** …, 2002 - academia.edu

… **management**. This paper describes the modeling of two fundamental crown fire **behavior** features: the onset of crowning and the **spread** … of the Canadian **Forest Fire Behavior Prediction** …

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The REBURN **model**: Simulating system-level forest succession and **wildfire dynamics**

SJ Prichard, RB Salter, PF Hessburg, NA Povak… - Fire Ecology, 2023 - Springer

… is the dominant driver of system-level vegetation **dynamics** within the Tripod landscape, the … REBURN **model** development and calibrated **model behavior**. We include a full **model** …

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Mapping **wildfire** susceptibility with the BURN-P3 **simulation model**

MA Parisien, VG Kafka, KG Hirsch, JB Todd, SG Lavoie… - 2005 - researchgate.net

… The **model** combines deterministic fire growth based on the Canadian Fire **Behavior Prediction** … probabilistic fire ignitions and **spread** events derived from historical fire and weather data. …

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Advanced automatic **wildfire** surveillance and monitoring network

D Stipaničev, M Štula, D Krstinić, L Šerić… - … on **Forest Fire** …, 2010 - academia.edu

… In this paper, after a short **survey** of various approaches to the **wildfire** monitoring, … **management** activities. GIS data is important for fire **behavior** modeling, **forest fire spread simulation** …

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